3/09/23

PHYSICS

1. Hypermetropia

2. ~~Repulsion~~ Repulsion

3. virtual and erect

4. below the apparent image of the fish

5. Given: $P = +1.5D$

$$f = \frac{1}{P}$$

$$f = \frac{1}{1.5} m$$

$$f = \frac{2}{3} m$$

$$= \frac{2}{3} \times 100$$

$$f = 66.7 \text{ cm}$$

The lens are converging, thus they are convex lens.

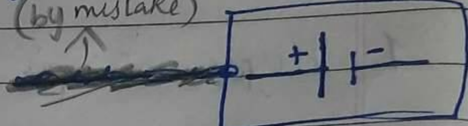
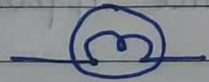
6. Current flowing (I) = 5 Amp
Time (t) = 30 mins

$$I = \frac{Q}{t}$$

$$Q = It$$
$$Q = 5 \times 1800$$
$$Q = 9000 C$$
$$\therefore Q = 9 \times 10^3 \text{ coulombs}$$

Ans: 9×10^3 coulombs of charges flows through the circuit

7.

Circuit Components	Symbol
1) Electric cell	(by mistake) 
2) Switch (in off position)	
3) Switch (in on position)	
4) Electric bulb	
5) Battery	

8.

$f = +12$ cm
$h = 5$ cm
$u = -20$ cm

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-20} = \frac{1}{12}$$

$$\frac{1}{v} + \frac{1}{20} = \frac{1}{12}$$

$$\frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{5-3}{60}$$

$$\frac{1}{v} = \frac{2}{60}$$

$$v = \frac{60}{2} = 30$$

2	20, 12
2	10, 6
5	5, 3
3	1, 3
	1

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$\boxed{v = 30\ cm}$

Image is formed 30 cm behind the convex lens. and the image is real and inverted.

9. The potential diffence between the two ends of the conductor (V) = ~~10~~ 10V - 5V = 5V

∴ potential difference = 5V

(W) Work done = 5 J
(Q) charge = ?

$$V = \frac{W}{Q}$$

$$Q = \frac{W}{V}$$

$$Q = \frac{5J}{5V}$$

$$\boxed{Q = 1\ coulomb}$$

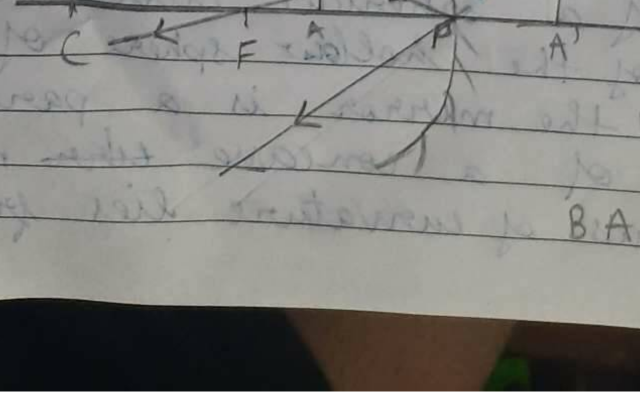
10.
a) Pole : The centre of a spherical mirror is called the pole. It lies in front of a concave mirror.

b) ~~Cor~~ Radius of curvature : Radius of curvature of a spherical mirror is the radius of the hollow sphere of glass of which the mirror is a part. In case of a concave ~~[illegible]~~ mirror the radius of curvature lies front of the mirror.

c) Principal ~~focus~~ axis: The staright line passing through the centre of curvature and pole of a spherical mirror is called its principal axis. The principal focus lies in front of ~~th~~ a concave mirror.

d) Principal ~~axis~~ focus: It is ~~a~~ the point of the principal axis at which the rays parallel to the principal axis meet after reflection. It is ~~reprented~~ represented by (F). The principal focus lies in front of the concave mirror.

10. The Object distance should be less than the focal length for the formation of an erect image ~~on~~ on a concave ~~mir~~ mirror. Hence, the range of distance of object from the mirror should be ~~les~~ less than 20 cm i.e. from 0 to 20 cm in front of the mirror from the pole. The nature of image so formed will be virtual and erect. The size of the image will be larger than the object.



B'A' - image of candle

B A - candle flame

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11. a) Since blue colour has a shorter wavelength and red colour has a longer wavelength, the red colour is ~~seen~~ able to reach our eyes after the atmospheric ~~st~~ scattering of light whereas the blue colour is scattered all over. Therefore, the sun ~~appere~~ appears reddish during early in the morning.

b) Power of accomodation of eye is the ability of the eye lens to focus near and far objects clearly on the retina by adjusting its focal length. ~~Power~~ Power of accomodation ~~is~~ of the eye is limited. It implies the focal length of the eye lens cannot be reduced beyond certain minimum limit.

c) Least distance of distinct vision for a normal human being is 25 cm.